RAPT: A TRANSFORMER-BASED APPROACH FOR ROBUST SYMBOLIC SCORE-TO-PERFORMANCE ALIGNMENT

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ABSTRACT

This study introduces RAPT (Robust Alignment via Piano Transformer), a score-to-performance framework for symbolic music alignment that combines Dynamic Time Warping (DTW) with learned semantic embeddings. The method acts as a contextual compass for the underlying DTW alignment: a lightweight Transformer encoder augments traditional piano roll and onset features with global structural representations to navigate macro-level ambiguities, while a DualDTW baseline resolves locally uncertain regions. Experimental results show that RAPT achieves competitive performance on standard benchmark datasets and remains stable in challenging settings involving performance errors such as wrong notes, hesitations, and local timing deviations. Unlike conventional methods that rely on frame-level similarity alone, RAPT's encoder is trained contrastively on aligned score-performance pairs. The embeddings encode the surrounding musical context of each frame, helping the alignment to distinguish between locally similar frames that occur in different parts of the piece. At inference time, the learned embeddings are fused with piano roll and onset features extracted from the score and the performance. The resulting feature sequences are then aligned using a DTW-based method, followed by additional refinement steps to resolve unmatched and uncertain regions.

1. INTRODUCTION

Score-to-performance alignment is a fundamental task in music information retrieval (MIR). Accurate alignments enable a wide range of downstream tasks, such as performance analysis [1] and automatic accompaniment [2]. However, while significant progress has been made in aligning polished performance, the task remains problematic when applied to the noisy data [3,4].

Human piano performances often deviate from the notated score through wrong, omitted, or repeated notes, as well as local timing irregularities such as hesitations and tempo instability [5]. Most note-level symbolic alignment methods rely on local pitch and onset features, whether through frame-level piano roll similarity [6] or event-level emission probabilities [7]. When several score locations share similar local pitch and onset patterns, these frame-level costs become ambiguous, and the warping path can drift away from the correct score position. Alignment errors are therefore often concentrated around regions that contain performance mistakes [6].

This work introduces **RAPT** (Robust Alignment via Piano Transformer), a framework for symbolic music alignment that augments Dynamic Time Warping (DTW) [8] with learned semantic embeddings. We conceptualize the method as a semantic compass for DTW: whereas conventional DTW relies primarily on local frame-level similarity, RAPT incorporates broader musical context through a lightweight Transformer encoder [9] operating on short temporal windows ($< 1\,\text{s}$). The resulting embeddings encode the surrounding musical context of each frame and provide a structural signal that helps the alignment distinguish between locally similar regions occurring in different parts of the piece.

The contributions of this paper are as follows:

1. We propose RAPT, a symbolic music alignment architecture that combines DTW-based alignment with context-aware semantic embeddings learned by a lightweight Transformer encoder.

2. We develop an alignment pipeline in which these embeddings complement piano roll and onset features, leading to more stable alignment in challenging regions affected by performance errors and local temporal irregularities. [1]

The rest of this paper is structured as follows: Section 2 presents an overview of symbolic music alignment. Section 3 introduces a detailed description of RAPT. Section 4 describes the experimental setup for the paper. Section 5 presents the results of the experiments and discusses the results. Finally, Section 6 concludes the paper.

2. RELATED WORK

This section provides a brief overview of symbolic score-to-performance alignment. For a general overview of music alignment, we refer the reader to [10].

[1] Supplementary Materials and the code to reproduce the experiments in this paper can be found in the following url https://ogh9hmhz60.github.io/RAPT-Alignment/.

Symbolic alignment was long dominated by Hidden Markov Model (HMM) formulations [11, 12]. The most established representative is Nakamura's et al.'s merged-output HMM model [7], which operates on event-level pitch sequences with a chord-level preprocessing step. Alignment is computed in two passes: a temporal HMM provides a preliminary alignment, after which performance errors are detected and a merged-output HMM with automatic hand separation is applied to local error regions.

More recent work has shifted toward Dynamic Time Warping with traditional low-level symbolic music representations (e.g., piano rolls and MIDI pitch events) [2, 13]. The *DualDTWMatcher* [13] represents one of the current state-of-the-art (SOTA) approaches for symbolic alignment. It is also event-based: a coarse DTW pass over pitch-set sequences yields an approximate score-to-performance mapping, after which a second DTW pass on per-pitch onset sequences extracts note-level matches. By separating pitch and time into two stages, DualDTWMatcher decouples the structural matching of pitch sets from the fine-grained alignment of onset times

Learned representations have only recently been integrated into symbolic alignment. *TheGlueNote* [14] is the closest neural predecessor of our work: a Transformer encoder predicts pairwise note similarities between two 512-note subsequences, trained with synthetic augmentations covering repeats, skips, insertions, deletions, and trills. The resulting similarity matrix is post-processed with a weighted DTW variant to obtain note matches.

3. RAPT

In this section we provide an overview of the RAPT system. We use FastDTW [15] for efficient alignment over our frame-level piano roll representation, augmented with learned semantic embeddings that encode *where* in a piece a given moment occurs. DTW therefore operates on a context-aware feature space combining structural and contextual information. The Transformer serves as a **contextual compass**, providing a high-level semantic signal that constrains the alignment. This prevents the warping path from drifting due to local ambiguities. Using the bidirectional context, the model remains robust to pitch mismatches and local tempo fluctuations, recognizing the correct structural position even when surface-level features are corrupted by performance errors. Figure 1 provides an overview of the training and inference pipeline.

3.1 Feature Representation

We represent the score and performance as feature matrices $\mathbf{S} \in \mathbb{R}^{T_s \times 89}$ and $\mathbf{P} \in \mathbb{R}^{T_p \times 89}$, where T denotes the number of frames. These are derived by parsing symbolic music files, MIDI or MusicXML, into quantized piano roll matrices using the `Partitura` library [16]. We adopt a constant frame rate F that determines the temporal quantization: F frames per second for performances (which use real time) and F frames per beat for scores (which use musical time). This produces aligned frame sequences that DTW can compare directly. This resolution is chosen to provide sufficient granularity for rapid ornaments (e.g., trills) while maintaining a manageable sequence length for the Transformer.

3.2 Encoder Architecture

We propose the **PianoTransformer**, a lightweight Transformer encoder [9] designed to extract context-aware embeddings from symbolic music. The model maps a local window of size W over the piano roll frames, centered at time t, to a single d-dimensional embedding vector $\mathbf{z}_t \in \mathbb{R}^d$. With our chosen configuration ($d = 256$, $L = 3$ layers, $H = 4$ heads), each PianoTransformer has 1.6M trainable parameters; the 4-seed ensemble used at inference totals 6.5M parameters.

The input frames are first linearly projected to the model dimension d. To encode temporal dependencies within the window without the rigid constraints of absolute positional encodings, we augment the sequence with relative positional information via a residual depthwise convolution. This specific technique, adapted from speech models [17, 18], allows the encoder to remain translation-invariant, making it more robust to the non-linear timing fluctuations typical of expressive piano performances.

The augmented sequence is processed by L Transformer encoder layers. Let $\mathbf{H}_t = (\mathbf{h}_t^{(1)}, \ldots, \mathbf{h}_t^{(W)}) \in \mathbb{R}^{W \times d}$ denote the output of the final encoder layer for the windowed input centered at time t. We extract the center index $c = \lfloor W/2 \rfloor$ using center pooling; alternative pooling strategies are evaluated in Section 4.4.3. We apply layer normalization (LN) and L2-normalization to produce the final embedding:

$$\mathbf{z}_t = \frac{\mathrm{LN}(\mathbf{h}_t^{(c)})}{\|\mathrm{LN}(\mathbf{h}_t^{(c)})\|_2}. \tag{1}$$

Center pooling, combined with the encoder's bidirectional self-attention, ensures that $\mathbf{z}_t$ incorporates information from the full temporal context surrounding the target frame. All architectural hyperparameters are detailed in Table 1 in the Supplementary Materials.

3.3 Contrastive Training

To ensure that score and performance representations are mapped into a shared geometric space, the encoder is trained as a weight-shared Siamese network [19]. Shared weights guarantee that identical musical structures in both domains yield comparable embeddings, which is a prerequisite for our distance-based alignment.

From ground-truth note alignments, we extract temporally corresponding score–performance window pairs. We optimize the InfoNCE loss [20] $\mathcal{L}$ (defined in Equation 2) over a mini-batch of B pairs to maximize the similarity of matched pairs while minimizing the similarity of mismatched ones:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(\mathbf{z}_i^s \cdot \mathbf{z}_i^p \,/\, \tau)}{\sum_{j=1}^{B} \exp(\mathbf{z}_i^s \cdot \mathbf{z}_j^p \,/\, \tau)}, \tag{2}$$

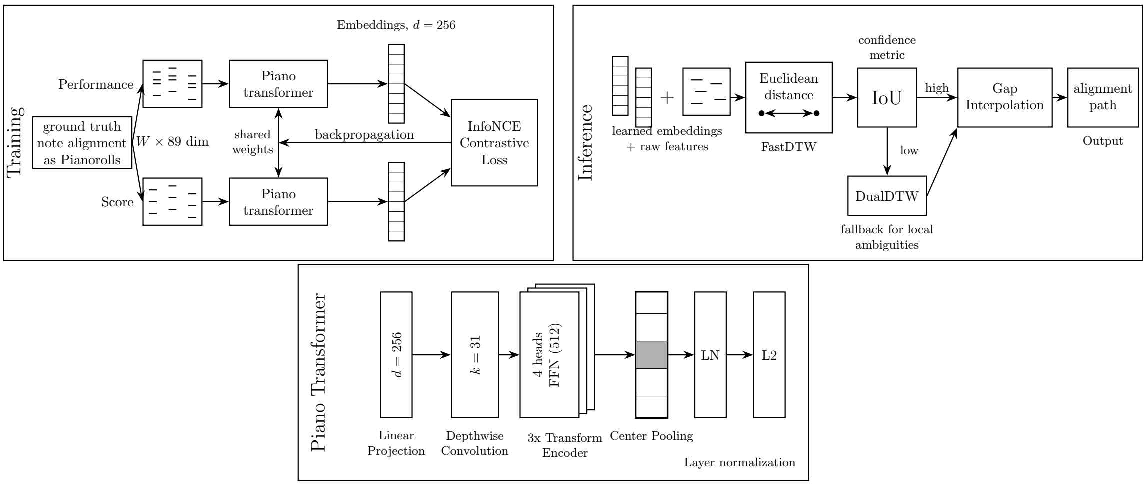

Figure 1. RAPT Architecture and pipeline overview: training, transformer, inference

where $\mathbf{z}_i^s$ and $\mathbf{z}_i^p$ denote the extracted embeddings for the i-th score and performance windows, respectively, and τ controls the temperature scaling.

During training, stochastic augmentations such as temporal jitter, note dropout, rubato simulation, pitch perturbation are applied to the performance branch only. This mirrors the real-world asymmetry where the notated score is fixed and only the human performance deviates.

3.4 Inference Pipeline

At inference, RAPT proceeds in five stages:

(1) Embedding extraction. Dense embedding matrices $\mathbf{E}_s$ and $\mathbf{E}_p$ are constructed by sliding the context window of size W across the score and performance sequences, extracting the respective $\mathbf{z}_t^s$ and $\mathbf{z}_t^p$ vectors at each frame. To mitigate the variance inherent to contrastive training, we ensemble multiple model checkpoints, trained with different random initialization seeds, by computing the element-wise mean of their output embeddings.

(2) Blended cost features. We construct a blended feature vector $\mathbf{f}_t^m$ for each frame t in both modalities ($m \in \{s, p\}$). The vector concatenates the *learned* ensembled Transformer embedding $\mathbf{e}_t^m$ with two *raw* structural features, the piano roll $\bar{\mathbf{r}}_t^m$ and the onset indicator $\mathbf{o}_t^m$, each weighted by an independently tuned scalar:

$$\mathbf{f}_t^m = \left[\, \alpha \cdot \mathbf{e}_t^m \ ; \ \beta \cdot \bar{\mathbf{r}}_t^m \ ; \ \beta \cdot \mathbf{o}_t^m \,\right], \tag{3}$$

where α controls the contribution of the Transformer's contextual representation and β controls the contribution of the structural piano roll and onset signals; both lie in $[0, 1]$ and are tuned per dataset.

(3) DTW and note matching. FastDTW [15] with Euclidean distance is applied to the blended features. We apply a Sakoe-Chiba band constraint (radius R) to restrict the search space, preventing the warping path from exploring pathologically large temporal deviations. To translate this frame-level path into one-to-one note matches, we project each score note's frame span through the path. We compute the Intersection over Union (IoU) of this projected span against all pitch-compatible performance notes. The performance note maximizing this IoU is selected, as it represents the highest temporal overlap and thus the most probable match.

Dataset	Pairs	Train	Val	Test
Vienna 4x22 [21]	88	63	17	8
Batik plays Mozart [22]	36	26	7	3
(n)ASAP [6]	833	584	166	83

Table 1. Datasets used in this work. (n)ASAP statistics refer to the robustly aligned subset only.

(4) Fusion with DualDTW. The IoU values from stage 3 serve as per-note confidence metrics. We retain matches where IoU $\geq \theta_{\text{keep}}$ (high-confidence). For unaligned notes in low-confidence regions (IoU $< \theta_{\text{keep}}$), the system falls back to the DualDTWMatcher [13] to resolve local ambiguities. Any performance notes that remain unmatched after this fusion are classified as *insertions* (i.e., extra notes played by the performer).

(5) Gap interpolation. Finally, we define the secure, high-confidence matches from stage 4 as *anchors*. We examine the unmatched sequences bounded by these anchors. If the temporal distance between two anchors is below the safety threshold Δ_{max}, unmatched score notes are greedily assigned to available, pitch-compatible performance candidates that strictly fall within the same intermediate index range.

4. EXPERIMENTAL EVALUATION

4.1 Datasets

We evaluate RAPT on three publicly available symbolic music alignment datasets, each providing pre-aligned score–performance pairs in MIDI format. These datasets are listed in Table 1.

Real piano performances contain deviations from the score: timing fluctuations, dropped notes, pitch errors, and hesitations. To evaluate alignment robustness under such conditions, we generate corrupted variants of each test performance using the *piano-synmist* framework [5], focusing on four error types that capture common forms of human deviation:

Mistouch errors insert short-duration notes near the target pitch, modeling accidental key presses.

Pitch Change errors replace a note with a different pitch while preserving timing, modeling pitch errors due to misreading or memory lapses.

Drag errors delay the onset of a note and propagate the timing shift to subsequent notes, modeling performer hesitations.

Forward/Backward Insertion errors insert notes from earlier or later in the performance into the current region, producing local note rearrangements.

For each test piece, we generate one corrupted variant per error type, yielding $4 \times N_{\text{test}}$ corrupted variants per dataset (with 5 (n)ASAP test pieces excluded due to failures in the corruption generation pipeline, producing 312 instead of 332 (n)ASAP variants).

4.2 Experimental Setup

Splits. For each dataset, we use fixed test sets of 8 (Vienna), 3 (Batik), and 83 ((n)ASAP) score–performance pairs, with the remaining pairs split into training and validation following an approximate 70/20 ratio of the total dataset. Exact partition sizes are reported in Table 1.

Training configuration. We train a 4-seed ensemble of the PianoTransformer (Section 3.2) on each dataset with the InfoNCE objective (Section 3.3). Models are trained for up to 200 epochs with early stopping based on validation loss. On Batik, early stopping consistently triggered within the first epoch due to the small training set, and we adopt 15 epochs as a canonical fixed budget. All other hyperparameters follow Table 1 in the Appendix in the Supplementary Materials.

Hyperparameter tuning. For each dataset, the inference-time hyperparameters α, β, and θ_{keep} are tuned via Optuna [23] with Tree-structured Parzen Estimator (TPE) sampling on the validation split.

Baselines. We compare RAPT against three established symbolic alignment methods spanning different methodological families: Nakamura's HMM-based system (referred to as Nakamura in the rest of the text) [7], the DTW-based DualDTWMatcher [6], and the deep-learning-based TheGlueNote [14]. All baselines are run with their default configurations on the same test sets.

Evaluation metric. We report the note-level F1 score, computed as $F_1 = 2PR/(P + R)$ where precision P and recall R are the fractions of correctly identified note matches relative to predicted and ground-truth alignment sets, respectively [6, 7].

4.3 Main Results

We evaluate all methods on the clean test split of each dataset and on corrupted variants.

4.3.1 Clean Performance

Overall, RAPT achieves the highest score on Vienna 4x22 ($F_1 = 0.992$, $n = 8$), exceeding all baseline methods. On (n)ASAP, RAPT performs competitively with DualDTW ($F_1 = 0.978$ vs. 0.980, $n = 83$) and substantially outperforms Nakamura ($+0.007$) and GlueNote ($+0.009$). On Batik plays Mozart, RAPT scores $F_1 = 0.988$ ($n = 3$), within a narrow band where all methods exceed 0.98 on this small test set with limited room for differentiation.

4.3.2 Robustness to Performance Errors

Table 2 reports F1 under corruption. RAPT scores highest on Vienna 4x22 ($F_1 = 0.769$, $n = 32$), tied with DualDTW and ahead of Nakamura ($+0.010$) and GlueNote ($+0.013$); on (n)ASAP ($n = 312$) it scores 0.977, competitive with DualDTW (0.978) and above Nakamura ($+0.008$) and GlueNote ($+0.010$); on Batik ($n = 12$) it scores 0.988, close to the baselines.

The degradation ($\Delta = F_1^{\text{corrupt}} - F_1^{\text{clean}}$) reveals the same pattern: RAPT shows no drop on (n)ASAP ($\Delta = 0.000$), significantly less than all baselines ($p < .001$), and ties the smallest drop on Batik ($\Delta = -0.001$). The large Vienna drop (≈ 0.22 F1 across all methods) reflects this benchmark's structural homogeneity and prevents differentiation between approaches.

4.4 Ablation Study

We ablate the major design choices of RAPT to quantify each component's contribution. The full table of results for the ablation can be found in Table 3 in the Supplementary Materials.

4.4.1 The Transformer Alone is Not Sufficient

Setting $\alpha = 1$, $\beta = 0$, and $\theta_{\text{keep}} = 0$ to rely on Transformer embeddings alone collapses F1 to 0.746 / 0.891 / 0.815 on Vienna / Batik / (n)ASAP. The DualDTW fallback and raw-feature blending together resolve a significant share of matches that the Transformer cannot place reliably.

4.4.2 Quality-Control Mechanisms

Accepting all Transformer matches without IoU filtering ($\theta_{\text{keep}} = 0$) drops Batik F1 by 0.010 ($0.988 \to 0.978$). Disabling gap interpolation changes F1 by at most ± 0.003 on clean and ± 0.0003 on corrupted data; the IoU-thresholded fusion already routes failure cases to DualDTW, leaving few gaps for interpolation to address.

4.4.3 Architectural Choices are Robust

Mean or max pooling, removing the onset channel, and single-seed inference each produce F1 within ± 0.003 of the canonical model (see Table 3 in the supplementary materials). The 4-seed ensemble yields small gains on Vienna ($+0.001$) and (n)ASAP ($+0.003$) with no change on Batik, confirming that alignment quality is driven by the hybrid pipeline rather than encoder details.

Method	**Vienna** ($n = 8$)			**Batik** ($n = 3$)			**ASAP** ($n = 83$)		
	Clean	Corrupt	$\Delta^{\dagger}$	Clean	Corrupt	$\Delta^{\dagger}$	Clean	Corrupt§	Δ
Nakamura	.984	.759	-.225	**.995**	**.992**	-.003	.971	.969	<u>-.001</u>
DualDTWMatcher	<u>.989</u>	<u>.769</u>	**-.220**	<u>.991</u>	<u>.989</u>	<u>-.001</u>	**.980**	**.978**	-.001
TheGlueNote	.976	.756	<u>-.221</u>	.980	.978	-.002	.969	.967	-.002
RAPT (Ours)	**.992**†	**.769**†	-.223	.988†	.988†	**-.001**	<u>.978</u>***	<u>.977</u>**	**.000**‡

*/**/***: Wilcoxon signed-rank vs. best baseline, Holm–Bonferroni corrected ($p<0.05/0.01/0.001$). ‡: RAPT degrades significantly less under corruption than best baseline (Wilcoxon on δ_i, corrected). †: $n \leq 8$; test is descriptive only. §: evaluated only on 78 pieces, as 5 pieces failed corruption

Table 2. Main alignment F1 results. Best score per column is **bold**, second best is <u>underlined</u>.

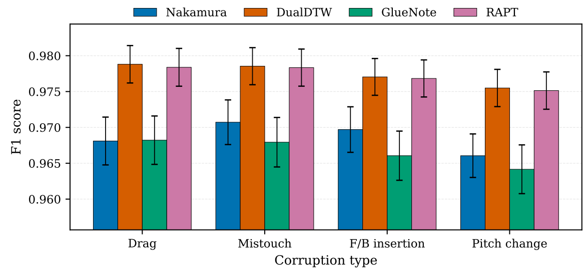

Figure 2. Mean F1 score per corruption type on (n)ASAP corrupted ($n = 78$ pieces per corruption type, $n = 312$ total). Error bars show the standard error of the mean. RAPT and DualDTW perform comparably across all four corruption types; both substantially outperform Nakamura and GlueNote, with the largest absolute gap on pitch changes and forward/backward insertions, where local-feature methods are most strongly affected by perturbations.

4.5 Cross-Corpus Generalization

To assess transfer across corpora, we train two joint ensembles, one on (n)ASAP+Batik, the other on (n)ASAP+Vienna and evaluate each on the held-out third corpus. Inference hyperparameters are tuned on the joint validation set. Cross-corpus RAPT matches DualDTWMatcher on both directions (within 0.001 F1) and exceeds Nakamura and TheGlueNote by 0.005–0.013, suggesting that the contrastively-learned embeddings transfer to corpora unseen during training.

5. DISCUSSION

5.1 Where RAPT's Contextual Embeddings Help

A per-corruption-type breakdown on (n)ASAP corrupted (Figure 2) shows that RAPT's largest advantages over Nakamura's HMM aligner [7] and GlueNote [14] are on pitch changes (+0.009 and +0.011 respectively) and forward or backward insertions (+0.007 and +0.011). These error types alter the local pitch and onset content of the performance, where frame-level approaches based on pitch–time emission probabilities or local feature similarity are most strongly affected. RAPT's contextual embeddings, learned via the InfoNCE objective described in Section 3.3, instead encode the broader musical surrounding of each frame and provide a structural signal that remains stable under local perturbations. This pattern is consistent with the design intent of using a Transformer encoder as a contextual compass for the alignment, analogous to context-aware encoders in speech recognition [17, 18].

5.2 Where the Transformer Compass Matters Most

To understand the role of the Transformer's contextual guidance beyond aggregate F1, we examine the per-piece distribution of test-set scores against all baselines. While aggregate F1 differences between RAPT and DualDTW are small on (n)ASAP and Batik (Section 4.3.1), the per-piece analysis reveals where the Transformer's contribution matters most.

On Vienna 4x22, RAPT achieves the highest F1 among all four methods on 4 of 8 test pieces, with the largest gains on Chopin Op. 38 (RAPT $F_1 = 1.000$ vs. worst baseline 0.971, $\Delta = +0.029$) and Schubert D783 pieces. On (n)ASAP, RAPT is the strict top performer on 11 of 83 pieces, with the most striking advantage on Schubert Impromptu Op. 90 D.899/1 (RAPT $F_1 = 0.957$ vs. worst baseline GlueNote 0.875; $\Delta = +0.083$). On the remaining pieces, RAPT and DualDTW perform comparably (median per-piece $|\Delta| < 0.001$), indicating that the hybrid pipeline matches DualDTW's strong baseline performance while providing additional headroom on the more difficult pieces.

Figure 3 illustrates this on Chopin Op. 38 performer 11 ($F_1 = 1.000$ vs. DualDTW 0.978): RAPT recovers all 16 of DualDTW's alignment errors, which cluster in the opening and final cadence.

These cases share a common signature: pieces with locally ambiguous pitch and onset content (e.g., Chopin's dense ornamentation and repetitive textures), where the Transformer's broader contextual signal helps the alignment maintain global structural correctness. Combined with the pure-Transformer ablation (Section 4.4.1) showing F1 collapse without the Transformer, this pattern suggests that the Transformer's contribution is not measured by its direct share of final matches, but by its ability to disambiguate alignment trajectories in regions where local features alone are insufficient.

5.3 Methods That Did Not Improve The System

Several expected improvements proved marginal. **Gap interpolation** produced F1 within ± 0.003 on clean and ± 0.0003 on corrupted data; the IoU-thresholded fusion routes failure cases to DualDTW directly, leaving few gaps for interpolation to address. **Synmist-augmented**

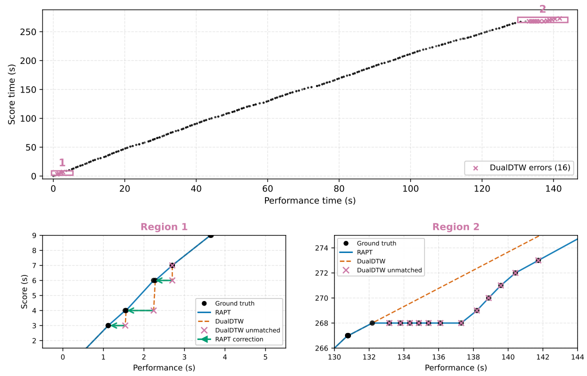

Figure 3. Per-note alignment errors on Vienna 4x22, Chopin op. 38 played by Pianist 11. Black: ground truth. Blue: RAPT. Orange dashed: DualDTW. **Region 1**: green arrows show RAPT correcting DualDTW's one-note shift. **Region 2**: DualDTW abandons the alignment in the final cadence; RAPT recovers all twelve unmatched notes.

training (fresh and warm-start variants) changed Vienna F1 by less than 0.001 on both clean and corrupted data, suggesting that static synthetic corruption does not provide additional signal beyond the contrastive objective. **Architectural variants** (mean/max pooling, 88-dim input, deeper encoders) similarly produced near-baseline F1 (Section 4.4.3), confirming that alignment quality is driven by the hybrid pipeline rather than encoder details.

5.4 Score-Level Diversity Drives Transformer Reliability

The pure-Transformer ablation (Section 4.4.1) revealed substantially larger F1 drops on Vienna (-0.246) than on Batik (-0.097) or (n)ASAP (-0.177), despite Vienna having a larger training set than Batik in score-performance pairs. This counterintuitive pattern, where the smaller dataset showed the smaller pure-Transformer drop, is consistent with the hypothesis that score-level diversity contributes more to the Transformer's reliability than raw training set size. With only 4 unique compositions in training, the encoder cannot learn structural patterns broad enough to generalize reliably without DualDTW backup.

6. CONCLUSIONS AND FUTURE WORK

We presented RAPT (Robust Alignment via Piano Transformer), a score-to-performance alignment system that combines a contrastively-trained Transformer encoder with a DualDTW fallback for low-confidence regions. Across three benchmarks (Vienna 4x22, Batik plays Mozart, and (n)ASAP), RAPT achieves competitive F1 with state-of-the-art alignment methods on clean data and demonstrates the smallest performance degradation under corruption on Batik and (n)ASAP. Ablations show that the alignment quality is determined primarily by the full pipeline rather than by individual architectural choices: pooling strategy, onset feature, and ensemble size each contribute marginally, while the IoU-thresholded fusion between Transformer and DualDTW is essential. The Transformer alone is insufficient on all three datasets, with the largest collapse on Vienna 4x22 despite its larger raw training set, suggesting that score-level diversity is a key driver of the encoder's reliability.

Several directions remain open: **On-the-fly augmentation with simulated mistakes.** We trained two variants of the Vienna ensemble on synmist-augmented data [5], both as a fresh training run and as a warm-start fine-tuning from the canonical model, but observed F1 changes within the noise floor of evaluation (Section 5.3). We hypothesize this is partly because the synthetic data was static: the model received finite exposure to a fixed set of corruption patterns rather than a continuous stream of varied perturbations. A more promising direction would be to integrate the mistake injector directly into the contrastive training loop, applying randomly-sampled corruption parameters on every training step, a standard recipe for improving robustness in other deep learning domains. **Improving alignment under heavy corruption.** On corrupted Vienna, all four methods including RAPT collapse to $\sim$0.77 F1 (Section 4.3.2), suggesting that current alignment systems struggle with heavily perturbed performances regardless of method. Two complementary directions could push this number higher. First, the contrastive objective could be extended to target robustness explicitly by mixing (clean score, corrupted performance) pairs into training, perhaps generated on-the-fly via the augmentation pipeline described above. Second, θ_{keep} could be made adaptive: rather than a fixed threshold, the system could detect heavy corruption and lower the threshold to retain more Transformer matches that would otherwise be over-rejected. We expect both directions would primarily benefit datasets with limited score-level diversity, where the encoder cannot rely on memorized structural patterns. **More compositions, or more performances?** The pure-Transformer ablation suggested that score-level diversity drives the Transformer's reliability more than raw training set size (Section 5.4). However, we cannot conclude this definitively from the existing benchmarks, where score diversity and performance count are confounded: Vienna provides 4 unique compositions performed by 22 pianists each, while Batik provides many more compositions but only one performance per piece. A controlled study, training on synthesized datasets where score diversity and performance count are varied independently, would be needed to confirm this interpretation. **Cross-corpus generalization.** The cross-corpus evaluation (Section 4.5) showed that joint training on two corpora produces an ensemble competitive with strong baselines on a held-out third corpus. Larger and more diverse training combinations, and corrupted cross-corpus settings, warrant further study. **Interpretability.** Visualization of the Transformer's learned representations, e.g., attention maps, t-SNE of frame-level embeddings, could illuminate the specific musical patterns the encoder captures. Such analysis would clarify how the model resolves alignment ambiguity in repeated regions and is a natural next step.

7. AI USAGE STATEMENT

AI-based writing assistance tools were used to improve the clarity, grammar and readability of selected parts of the manuscript. Specifically, we used Anthropic's Claude (Opus 4.6 and 4.7), ChatGPT 5.3 by OpenAI and Google's Gemini Pro 3.1. These tools were not used to generate experimental results, perform data analysis, or draw scientific conclusions. All AI-assisted content was reviewed and edited by the authors, who take full responsibility for the final manuscript.

8. ETHICS STATEMENT

This work uses publicly available musical material, and the score material considered in our experiments is based on compositions in the public domain. The study does not involve human subjects, personal data, biometric information, or user-identifying metadata. Our experiments focus on computational score-performance alignment and are not intended to evaluate individual performers or assess musical ability.

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